

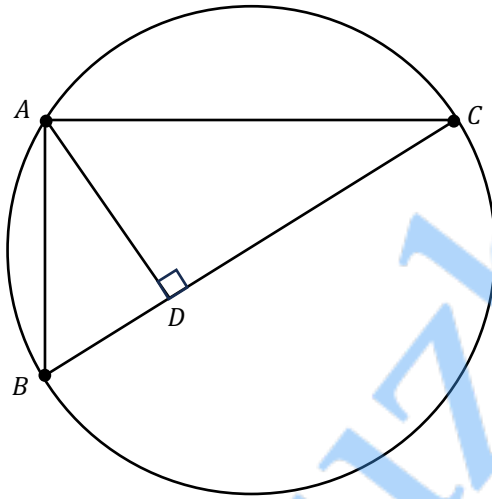
SAT UZBEKISTAN

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1. There are two ranches. Ranch A and ranch B . The ratio of male horses to female horses is 1: 15 for the total number of horses on ranch A and B . The ratio of female horses on ranch A to ranch B is 7: 17. There are 168 horses on ranch A and 344 horses on Ranch B . How many more male horses are there on ranch A than on ranch B ?

A) 24 B) 6 C) 8 D) 10

2. In the given figure, BC is the diameter of the circle. If the length of BC is equal to 132 and the length of AB is equal to $\sqrt{363}$, what is the value of $\frac{BC}{BD}$?



Note: Figure not drawn to scale.

3. A certain bacterial colony currently has 5,000 spores. If the number of spores in this colony quadruples every 33 hours, which of the following functions N gives the number of spores in the colony d days from now?

A) $N(d) = 5,000(4)^{\frac{11}{8}d}$
 B) $N(d) = 5,000(4)^{\frac{8}{11}d}$
 C) $N(d) = 5,000(4)^{\frac{5}{4}d}$
 D) $N(d) = 5,000(4)^{\frac{4}{5}d}$

4. In the given equations, a and b are constants, $a > 1$ and $b > 1$. What is the value of x ?

$$\sqrt[7]{a^5} = \sqrt{b^5}$$

$$a^{3x-3} = b^3$$

A) 27/21 B) 21/27 C) 32/21 D) 38/21

5. a is 3700% greater than b . a is also 5% less than c . What percent greater than c is b ?

A) 3900 B) 2600 C) 5200 D) 6500

6. The proposal for a development project was included on a city's election ballots. A local tv station reported that 6 times as many people voted in favor of the proposal as people who voted against it. If 1,770 more people voted in favor of the proposal than those who voted against the proposal. How many people voted against the proposal?

A) 354 B) 384 C) 324 D) 364

7. The table gives the volume and surface areas of two similar rectangles, where k is a constant.

	Volume (cubic inches)	Surface Area (square inches)
Rectangle A	4,374	1,782
Rectangle B	1,500,282	k

What is the value of k ?

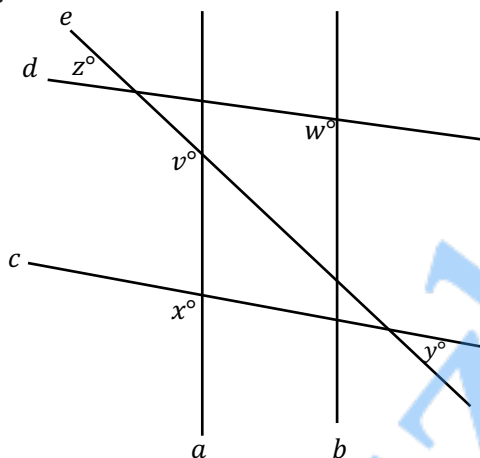
A) 5,346
 B) 12,474
 C) 87,318
 D) 611,226

8. In the given system of equations, c is a constant. The system has two distinct real solutions. Which of the following could be the value of c ?

$$\begin{aligned} y &= x - c \\ y &= -6(x - 5)^2 \end{aligned}$$

- A) 2
B) 4
C) $\frac{119}{24}$
D) 7

9.



Note: Figure not drawn to scale.

In the figure, parallel lines a and b are intersected by lines c , d , and e . If $x = 54$, $v = 127$, and $w > x$, which statement about y and z must be true?

- A) $z > y$
B) $z < y$
C) $z = y$
D) $z + y = 90$

10. An equilateral triangle has a perimeter of 1200 meters. If the height of the triangle is equal to $x\sqrt{3}$, what is the value of x ?

- A) 200 B) 400 C) 100 D) 250

$$f(x) = (x + a)(x + b)$$

11. For the given function $f(x)$, where a and b are integer constants, $f(-21) > 0$, $f(-15) > 0$, and $f(-18) < 0$, what is one possible value of $a + b$?

- A) 35 B) 39 C) 41 D) 43

12. Which system of linear equations has no solution?

- A) $\begin{aligned} -3x + 7y &= -9 \\ 3x - 7y &= 9 \end{aligned}$
B) $\begin{aligned} 3x - 7y &= 5 \\ 2x + 8y &= 6 \end{aligned}$
C) $\begin{aligned} 4x - 9y &= 7 \\ -12x + 27y &= -21 \end{aligned}$
D) $\begin{aligned} -3x + 8y &= 20 \\ 6x - 16y &= 40 \end{aligned}$

13. Function f is a quadratic function. The graph of $y = f(x)$ in the xy -plane has a vertex at $(-7, 14)$ and passes through the point $(-4, -31)$ and has a y -intercept at $(0, a)$. The graph of $3f(x)$ has a y -intercept at $(0, b)$. What is the positive difference between a and b ?

- A) 462 B) 465 C) 472 D) 476

14. Each of the following frequency tables represents a data set. Which of these frequency tables represent the data set with the smallest standard deviation?

A)

Value	Frequency
2	1
10	5
15	3
20	5
24	1

B)

Value	Frequency
2	0
10	20
15	25
20	20
24	0

C)

Value	Frequency
2	100
10	50
15	25
20	0
24	25

D)

Value	Frequency
2	45
10	55
15	65
20	75
24	85

15. In the given equation, k is a positive constant. The product of the solutions to the equation is 154. What is the value of k ?

$$\frac{3}{5}(5x + 11)(x + \sqrt{5k + 10})(x - \sqrt{5k + 10}) = 0$$

- A) 12 B) 15 C) 18 D) 21

16. Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(7x - 210)^\circ$. Which of the following could NOT be the sum of the measures of any two of these angles?

- A) $(-14x + 780)^\circ$
 B) $(-14x + 420)^\circ$
 C) 180°
 D) $(14x - 420)^\circ$

$$\sqrt{a - x} = 46 - x$$

17. In the given equation, a is a constant. The equation has exactly one real solution. What is the minimum possible value of $4a$?

- A) 83 B) 86 C) 82 D) 80

18. The mass of particles A , B , and C are a atomic mass units, b atomic mass units, and c atomic mass units, respectively. If the mass of particle A is 4,600% of the mass of particle C , and the mass of particle C is 0.004% of the mass of particle B , which expression represents the value of $a + b$ in terms of c ?

- A) $2,960c$
 B) $4,604c$
 C) $4,850c$
 D) $25,046c$

19. If $\frac{x-7}{5} = \frac{x-4}{3}$, the value $x - 4$ is between which of the following pairs of values?

- A) -9 and -4
 B) -1 and 1
 C) 2 and 5
 D) 7 and 11

20. A circle with a center $B(0, 0)$ is a unit circle on the xy -plane. Point $A(1, 0)$ and C lie on the circle. If the measurement of angle ABC is $\frac{975\pi}{60}$ in radians, what is the x -coordinate of point C ?
- A) $\frac{\sqrt{2}}{2}$
 B) $-\frac{\sqrt{2}}{2}$
 C) 0
 D) $\frac{\sqrt{2}}{3}$
21. A handyman charges a flat rate of \$218 for the first 3 hours of work and \$68 for each additional hour of work. Which equation gives the total amount y , in dollars, that the handyman charges for x hours of work, where $x > 3$?
- A) $y = 68x + 218$
 B) $y = 218x + 68$
 C) $y = 68x + 14$
 D) $y = 68x + 422$
22. For a certain circuit, its power P , in watts; current C , in amperes; voltage V , in volts; and resistance R , in ohms, are related as $\frac{CV^3}{P} = \sqrt{PR}$, where P, C, V , and R are positive. When $R = 34$, which equation correctly expresses P in terms of C and V ?
- A) $P = \frac{CV^3}{\sqrt{34}}$
 B) $P = \frac{CV^3}{\sqrt{34P}}$
 C) $P = \sqrt[3]{\frac{34}{C^2V^4}}$
 D) $P = \sqrt[3]{\frac{C^2V^6}{34}}$
23. $f(x) = x + 7$
 $g(x) = 5x^2 - kx + 245$
- The functions f and g are given. In function g , k is a constant. If $f(x) \cdot g(x) = 5x^3 + 1,715$, what is the value of k ?
- A) 35 B) 38 C) 41 D) 43
24. A circle is inscribed inside of square $ABCD$. Segment AC is the diagonal of the square, and its length is 30cm. What is the radius of the circle in cm?
- A) $\frac{15\sqrt{2}}{2}$
 B) 15
 C) $15\sqrt{2}$
 D) 30
25. Which of the following has $x + 2a$ as a factor where a is a positive integer?
- A) $3x^2 + 7x + 14a = 0$
 B) $3x^2 + 28x + 14a = 0$
 C) $3x^2 + 42x + 14a = 0$
 D) $3x^2 + 49x + 14a = 0$
26. A metal mold in the shape of a rectangular prism has a length of 27 inches, a width of 48 inches, and a height of 180 inches. The equation $c = v(d)$ calculates the cost of filling the metal mold, where c is the total cost in dollars, v is the volume in cubic yards, and d is the cost per cubic yard. If c is equal to \$970, what is the value of d in dollars? (Note 1 yard = 36 inches)
- A) 194 B) 196 C) 198 D) 200

27. The expression $\frac{x^{27}(x-7)}{3x^3} + \frac{7x^{27}}{3x^3}$ is equivalent to $\frac{1}{3}x^k$, where k is a constant and $x > 0$. What is the value of k ?
- A) 3
B) $\frac{27}{3}$
C) 25
D) 28
28. Megan engages in up to 3 types of exercise each week for a total of 14 hours while training for an ironman competition. Megan runs the same number of minutes each week. The equation $y = 780 - x - 200$ represents the situation where Megan swims for x minutes during a week and bikes for any remaining training time y , in minutes. If this equation is graphed in the xy -plane, which of the following statements is true?
- A) During a week when Megan runs for 780 minutes, she bikes for 200 minutes.
B) Each week, Megan swims and bikes for a total of 780 minutes.
C) During a week when Megan doesn't swim, she runs for 260 minutes.
D) During a week when Megan doesn't bike, she swims for 390 minutes.
29. The functions f and g are defined by the equations shown, where a and b are integer constants, $a > b$ and $b > 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or a coefficient, the maximum of the graph of the corresponding function when $x \geq 0$.
- I. $f(x) = b(0.97)^{x+a}$
II. $g(x) = b(0.97)^x + a$
- A) I only
B) II only
C) I and II
D) Neither I nor II
30. An equilateral triangle is inscribed in a circle with a diameter of 16. Which of the following gives the area of the equilateral triangle?
- A) $6\sqrt{3}$
B) $24\sqrt{3}$
C) $32\sqrt{3}$
D) $48\sqrt{3}$
31. The given equation $ax^2 + 88x + c = 0$, where a and c are constants has no real solutions and a factor of $kx + j$. What is the least possible integer value of ac ?
- A) 1937 B) 1441 C) 1446 D) 1451
32. $f(t) = 34,000(1.07)^{6t}$
- The given function f models the balance of an investment account, in dollars, t years after it is opened. Which statement is the best interpretation of $(1.07)^{6t}$?
- A) Every 6 years, the balance increases by 2,380 dollars.
B) Every 6 years, the balance increases by 7% of the previous 6 years balance.
C) Every 2 months, the balance increases by 2,380 dollars.
D) Every 2 months, the balance increases by 7% of the previous 2 months balance.
33. The quadratic function $a(x + 4.5)^2 - d$ can be rewritten as $(x - 9.5)(x + c)$. If a is equal to 1, what is the value of d ?
- A) 196 B) 198 C) 200 D) 202

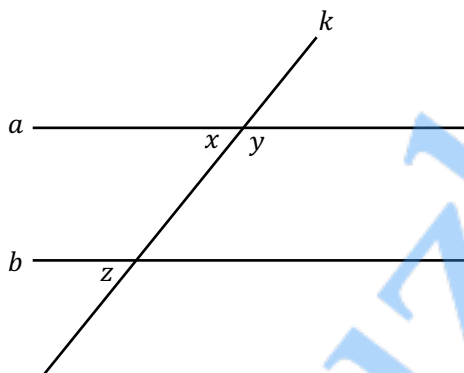
34. $g(x)$ is a quadratic function. In the xy -plane, the graph of $y = g(x)$ passes through the points $(12, 540)$ and $(16, 908)$. If $g(0) = 12$, what is the value of $g(24)$?

A) 1932 B) 1936 C) 1938 D) 1940

35. The surface area of rectangular prism A is 312 m^2 . The surface area of rectangular prism B is $11,232 \text{ m}^2$. Rectangular prism A has a volume of 224 m^3 . If both rectangular prisms are similar, what is the volume of rectangular prism B ?

A) 48324 B) 46324 C) 44324 D) 49324

36.



Note: Figure not drawn to scale.

The given figure shows lines a and b intersected by line k . Which of the following is sufficient to prove that a and b are parallel?

- A) $180 - y = x$
 B) $x = 180 - z$
 C) $y + z = 180 - x$
 D) $180 - z = y$

37. The function f is defined by $f(x) = ax^2 + bx + c$, where a , b and c are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(13, 0)$ and $(-6, 0)$. Which of the following is the value of $a + b$ in terms of a ?

- A) $7a$
 B) $6a$
 C) $-6a$
 D) $-7a$

38.

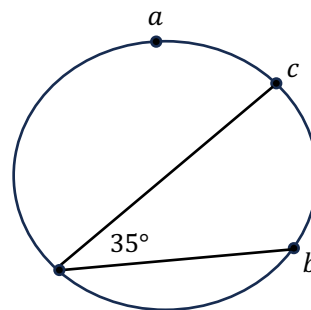
$$f(x) = 1.6^x + 6$$

$$g(x) = 1.8x + b$$

The given system of equations intersects at points (j, k) and (h, r) , where $r < k$. What is the least possible value of b ?

- A) 7
 B) 6
 C) 5
 D) 4

39.



For the given circle, the measure of arc $abc = 130 + 7x$ degrees and the measure of arc $acb = 90 - 3x$ degrees. What is the value of x ?

- A) 52.5 B) 54.5 C) 56.5 D) 58.5

40. The function f is defined by $f(x) = 78(0.39)^x$. For any positive integer n , the value of $f(n)$ is $p\%$ less than the value of $f(n - 1)$. What is the value of p ?

A) 78
B) 61
C) 39
D) 22

41. For which value of a does the quadratic $(7x - 8)^2 + 9 = a$ only have one solution?

A) 9 B) 12 C) 15 D) 18

42. $f(x) = a^x - b$

The exponential function given above passes through points $(c, 12)$ and $(2c, 768)$. What is a possible value of b ?

A) 16 B) 20 C) 24 D) 26

43. A 100 gram metal alloy is 40% silver. It contains a metal alloy that is 65% silver and a second metal alloy made up of 25% silver. What is the mass of the first metal alloy?

A) 37.5 B) 32.5 C) 42.5 D) 45.5

44. Joshua Tree National Park has an area of 1,242 square miles. What is the area, in square yards, of Joshua Tree National Park? (1 mile = 1,760 yards).

A) $\frac{1,242}{1,760}$
B) $\frac{1,242}{(1,760)^2}$
C) $1,242(1,760)$
D) $1,242(1,760)^2$